

FUNCTIONAL REQUIREMENTS DOCUMENT

Medical Appointment No-Show Prediction System

Document Type	Functional Requirements Document (FRD)
Related Documents	PRD v1.0, TRD v1.0
Version	1.0 (Draft)
Status	Proposed
Owner	Product / Business Analysis Team

1. Introduction

This Functional Requirements Document (FRD) describes, in detail, what the Medical Appointment No-Show Prediction System must do from a user and business-process perspective. It expands on the goals defined in the PRD and complements the technical design in the TRD by specifying functional behavior, user roles, workflows, business rules, and acceptance criteria.

2. Scope

This document covers the functional behavior of five system modules: Data Management, Prediction Engine, Risk Dashboard, Reporting & Analytics, and User & Access Management. It does not define infrastructure, deployment, or non-functional performance targets — see the TRD for those.

3. User Roles and Permissions

Role	Permissions
Clinic Administrator	Full access: manage users, view all dashboards/reports, configure risk thresholds
Front-desk / Scheduling Staff	View risk-ranked appointment lists, log outreach/contact outcomes
Doctor / Care Provider	View aggregate no-show risk for their own schedule (read-only)
Data/ML Analyst	Trigger retraining, view model performance reports, manage the data pipeline

4. Functional Requirements

Requirements are grouped by module. Each requirement has a unique ID and a priority (High / Medium / Low) reflecting its importance to the initial release.

4.1 Module: Data Management

ID	Requirement	Priority
FR-DM-01	The system shall allow authorized users to upload historical appointment data files (CSV) for training.	High
FR-DM-02	The system shall validate uploaded data against the required schema and report validation errors.	High
FR-DM-03	The system shall deduplicate and clean ingested records automatically before use.	High
FR-DM-04	The system shall log every data ingestion event with timestamp, record count, and error count.	Medium
FR-DM-05	The system shall support scheduled, recurring ingestion from a connected source system.	Low

4.2 Module: Prediction Engine

ID	Requirement	Priority
FR-PE-01	The system shall compute a no-show probability (0–100%) for every upcoming appointment.	High
FR-PE-02	The system shall classify each appointment into a risk tier (Low / Medium / High) using configurable thresholds.	High
FR-PE-03	The system shall re-score all upcoming appointments on a scheduled basis (e.g., nightly).	High
FR-PE-04	The system shall allow an authorized analyst to trigger model retraining on demand.	Medium
FR-PE-05	The system shall record which model version produced each prediction.	Medium
FR-PE-06	The system shall compare a newly trained model against the current production model before allowing promotion.	Medium

4.3 Module: Risk Dashboard

ID	Requirement	Priority
FR-RD-01	The system shall display a list of upcoming appointments ranked by no-show risk.	High
FR-RD-02	The system shall allow staff to filter the list by date, risk tier, and doctor.	High
FR-RD-03	The system shall let staff mark an appointment as "contacted" and log the outcome (confirmed, rescheduled, no response).	High
FR-RD-04	The system shall visually distinguish High/Medium/Low risk appointments (e.g., color coding).	Medium
FR-RD-05	The system shall allow doctors to view a read-only summary of their day's risk distribution.	Low

4.4 Module: Reporting and Analytics

ID	Requirement	Priority
FR-RA-01	The system shall display overall no-show rate trends over a selectable date range.	High
FR-RA-02	The system shall display no-show rate broken down by age group, gender, weekday, and SMS reminder status.	High
FR-RA-03	The system shall display current model performance metrics (Accuracy, Precision, Recall, F1, ROC-AUC).	Medium
FR-RA-04	The system shall allow administrators to export report data (CSV or PDF).	Medium
FR-RA-05	The system shall show intervention effectiveness (no-show rate for contacted vs. non-contacted high-risk patients).	Medium

4.5 Module: User and Access Management

ID	Requirement	Priority
FR-UA-01	The system shall require authentication for all users before granting access.	High
FR-UA-02	The system shall enforce role-based access control per Section 3.	High
FR-UA-03	The system shall allow administrators to create, edit, and deactivate user accounts.	Medium
FR-UA-04	The system shall log user actions relevant to patient data access for audit purposes.	Medium

5. Use Cases

UC-01: View Risk-Ranked Appointment List

Actor: Front-desk / Scheduling Staff

Preconditions: User is authenticated and appointments have been scored.

Main Flow:

- 1 User logs into the dashboard.
- 2 System displays upcoming appointments sorted by no-show risk (highest first).
- 3 User applies filters (date range, doctor, risk tier) as needed.
- 4 System updates the list based on selected filters.

Postconditions: User has a prioritized view of appointments to follow up on.

Exceptions: If no appointments are scored yet, the system displays a message indicating scoring is pending.

UC-02: Contact a High-Risk Patient and Log Outcome

Actor: Front-desk / Scheduling Staff

Preconditions: A high-risk appointment is visible in the dashboard.

Main Flow:

- 1 User selects a high-risk appointment from the list.
- 2 User contacts the patient (call/SMS) outside the system or via an integrated channel.
- 3 User records the outcome (confirmed, rescheduled, no response) in the system.
- 4 System timestamps and stores the outcome against the appointment record.

Postconditions: The appointment's contact status and outcome are saved and reflected in reporting.

Exceptions: If the appointment no longer exists (e.g., already rescheduled), the system notifies the user and refreshes the list.

UC-03: Trigger Model Retraining

Actor: Data/ML Analyst

Preconditions: New historical data has been ingested since the last training run.

Main Flow:

- 1 User navigates to the model management screen.
- 2 User initiates a retraining run.
- 3 System trains candidate models and evaluates them against a holdout set.
- 4 System presents comparison metrics against the current production model.
- 5 User approves or rejects promotion of the new model to production.

Postconditions: If approved, the new model becomes the active scoring model; if rejected, the current model remains active.

Exceptions: If the new model underperforms defined minimum thresholds, the system blocks promotion automatically.

UC-04: View No-Show Analytics

Actor: Clinic Administrator

Preconditions: Historical and current appointment/prediction data exists.

Main Flow:

- 1 User opens the Reporting & Analytics screen.
- 2 User selects a date range and optional breakdown (age, gender, weekday, SMS status).
- 3 System renders the corresponding charts and summary statistics.
- 4 User optionally exports the report.

Postconditions: User gains insight into no-show patterns to inform scheduling and outreach policy.

Exceptions: If no data exists for the selected range, the system displays an empty-state message.

6. Staff Workflow

The diagram below illustrates the typical day-to-day workflow for front-desk staff using the system.

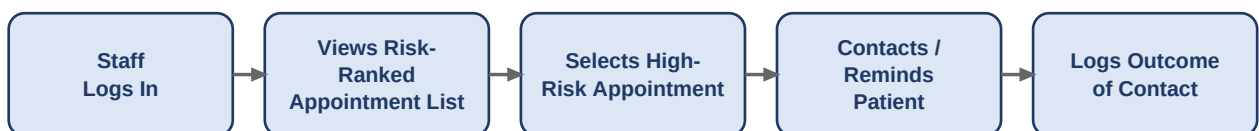


Figure 1. Staff daily workflow for managing high-risk appointments

7. Business Rules

- An appointment is classified High risk if no_show_probability ≥ 0.7 , Medium if between 0.4 and 0.7, and Low if below 0.4 (thresholds configurable by administrators).
- Only appointments at least 24 hours in the future are eligible for staff outreach actions.
- A patient's prior_noshow_rate is recalculated after every completed appointment (attended or not).
- Model retraining requires at least one full month of new confirmed appointment outcomes since the last training run.
- A new model can only be promoted to production if its Recall and F1-score are equal to or better than the current production model's, per FR-PE-06.
- All actions that access identifiable patient data must be attributable to an authenticated user account.

8. Data Validation Rules

Field	Validation Rule
Age	Must be a non-negative integer, reasonable upper bound (e.g., ≤ 120)
Scheduled Date / Appointment Date	Must be valid dates; Appointment Date $\geq$ Scheduled Date
Gender	Must match an accepted set of values
SMS Received	Boolean only
Appointment ID	Must be unique per record
No-Show (target, training data only)	Boolean only; required for training records

9. Assumptions and Dependencies

- Historical appointment data with a labeled No-Show outcome is available for initial training.
- Staff have basic familiarity with dashboard-style web applications.
- The system depends on the Prediction Engine (per the TRD) being available to serve scores before the dashboard can display risk rankings.
- Outreach (calls/SMS) is performed by staff outside the system in v1; the system only records outcomes.

10. Out of Scope

- Automated SMS/call dispatch (staff perform outreach manually in v1).
- Direct patient-facing features (e.g., patient self-service rescheduling).
- Mobile native application (web dashboard only in v1).

11. Glossary

Term	Definition
No-Show	A scheduled appointment the patient did not attend and did not cancel in advance
Risk Tier	A categorical label (Low/Medium/High) derived from the no-show probability
Model Registry	A store of versioned trained models and their evaluation metadata
Risk-Ranked List	The list of upcoming appointments ordered by descending no-show probability